AN ELEMENT AVOIDING THE MINIMAL PRIMES EXTENDS TO A SYSTEM OF PARAMETERS STABLE UNDER SMALL PERTURBATIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes '(f, g)' for the ideal (f, g_1, ..., g_{d-1}) in one sentence inside the counted proof span; the abbreviation is reproduced unchanged. The two references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Existence of a system of parameters extending a given element avoiding the minimal primes. invoked at proof lines 38363-38365. Supplies the g_1, ..., g_{d-1} that the perturbation argument needs.
- In a Noetherian ring a power of the radical of an ideal lies in the ideal. invoked at proof lines 38367-38368. Gives the exponent n from which N = n + 1 is built.
- The relation 1 + a is a unit when a lies in the maximal ideal. used at proof line 38381. Is what makes the first ideal generator absorb the perturbation without changing the ideal.

Source reference. Original lemma 00IM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 684–684. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 38086: sources/stacks-more-algebra/pinned-original-source.tex, source lines 38086–38086. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $(R, \mathfrak{m})$ be a Noetherian local ring of dimension d, and let $f \in \mathfrak{m}$ be an element not contained in any minimal prime ideal of R. Then there exist elements $g_1, \ldots, g_{d-1} \in \mathfrak{m}$ and $N \in \mathbf{N}$ such that*

(1) *$f, g_1, \ldots, g_{d-1}$ form a system of parameters for R*

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(2) *If $h \in \mathfrak{m}^N$, then $f + h, g_1, \ldots, g_{d-1}$ is a system of parameters and we have $\mathit{length}_R R/(f, g_1, \ldots, g_{d-1}) = \mathit{length}_R R/(f + h, g_1, \ldots, g_{d-1})$.*

Proof. By Lemma 38086 there exist $g_1, \ldots, g_{d-1} \in \mathfrak{m}$ such that $f, g_1, \ldots, g_{d-1}$ is a system of parameters for R. Then $\mathfrak{m} = \sqrt{(f, g_1, \ldots, g_{d-1})}$. Thus there exists an n such that $\mathfrak{m}^n \subset (f, g)$, see Algebra, Lemma 00IM. We claim that $N = n + 1$ works. Namely, let $h \in \mathfrak{m}^N$. By our choice of N we can write $h = af + \sum b_i g_i$ with $a, b_i \in \mathfrak{m}$. Thus

$$\begin{aligned}(f + h, g_1, \ldots, g_{d-1}) &= (f + af + \sum b_i g_i, g_1, \ldots, g_{d-1}) \\ &= ((1 + a)f, g_1, \ldots, g_{d-1}) \\ &= (f, g_1, \ldots, g_{d-1})\end{aligned}$$

because $1 + a$ is a unit in R. This proves the equality of lengths and the fact that $f + h, g_1, \ldots, g_{d-1}$ is a system of parameters. □